Task 4 Question

How would you redesign the US rule pack to reflect the actual current US legal reality (intent-based discrimination only, no disparate impact) while keeping the EU mode focused on outcome-based bias testing under the AI Act without simply turning off all fairness monitoring in the US?

Short Answer

I would redesign the tool by separating legal compliance triggers from internal fairness monitoring signals. The US rule pack should no longer treat a low disparate-impact ratio as a direct ECOA/Regulation B violation or as an automatic US legal failure. Instead, US mode should focus its legal findings on direct use of prohibited-basis attributes, differential treatment of similarly situated applicants, suspicious human-override patterns, prohibited-basis discouragement, inaccurate adverse-action explanations, and proxy-risk indicators where there is evidence suggesting intentional or pretextual discrimination. Outcome metrics should still be calculated, but they should be labelled as monitoring-only governance indicators, not legal liability tests.

The EU rule pack should remain outcome-focused because creditworthiness and credit-score AI systems are classified as high-risk AI systems under the AI Act. For the EU, approval-rate disparities, false-negative-rate gaps and other group outcome metrics should remain part of the high-risk AI governance evidence package, alongside technical documentation, logging, human oversight, data governance, risk management and lifecycle monitoring.

Legal and Regulatory Basis for the Redesign

US basis. The redesigned US pack should align with the CFPB final rule position that ECOA does not authorize disparate-impact liability, also called the "effects test". The final rule states that ECOA does not authorize disparate-impact liability and further narrows the role of effects-based analysis in Regulation B. It also states that the effective date is 90 days after Federal Register publication. Therefore, a low approval-rate ratio should not be presented as a US legal violation by itself. However, Regulation B still prohibits discrimination on a prohibited basis in any aspect of a credit transaction, and the official interpretation continues to recognize disparate treatment, including situations where similarly situated applicants are treated differently without a legitimate nondiscriminatory reason or where the stated reason is pretextual.

Adverse-action explainability. The US pack should also retain a strong adverse-action module. The CFPB has stated that ECOA and Regulation B require creditors to provide specific reasons to applicants against whom adverse action is taken, and that this requirement applies equally to decisions made using complex algorithms. A creditor cannot rely on model complexity as a reason for failing to give accurate and specific reasons.

EU basis. For the EU pack, the AI Act Service Desk identifies AI systems used to evaluate the creditworthiness of natural persons or establish their credit score as high-risk systems, except where used for detecting financial fraud. The European Commission describes the AI Act as a risk-based framework intended to protect safety, fundamental rights and human-centric AI. Therefore, in EU mode, outcome-based bias testing remains appropriate as part of high-risk AI risk management and evidence generation.

What Is Wrong with the Current Prototype Design?

The current prototype is useful as a classroom demonstration, but the US and EU rule packs are too similar. Both modes use a disparate-impact threshold: 0.80 in US mode and 0.90 in EU mode. Both can return REVIEW_REQUIRED because the same group disparity falls below the selected threshold. That makes the US pack look like a lighter version of the EU pack rather than a truly jurisdiction-aware legal design. The revised design should keep the same fairness calculations but change how the results are interpreted.

Redesigned Compliance Logic

Design element	US redesigned rule pack	EU rule pack
Legal theory	Intent-based / disparate-treatment logic. No automatic ECOA liability from disparate impact alone.	High-risk AI governance. Outcome-based bias testing remains part of risk management evidence.
Role of disparate-impact ratio	Monitoring-only risk signal, not a legal violation trigger.	Compliance-relevant bias metric and governance review trigger.
Primary legal checks	Direct protected-attribute use; similarly situated treatment; override disparities; discouragement; adverse-action reason accuracy; proxy risk with intent/pretext indicators.	Bias testing; data governance; technical documentation; audit logs; human oversight; lifecycle monitoring; robustness and accuracy controls.
Output label	No US legal finding / legal review required / monitoring review recommended.	High-risk governance review required / governance controls complete.
Explanation to user	“Disparity does not by itself establish a US ECOA violation; it is an internal	“Disparity forms part of high-risk AI evidence and may require mitigation, documentation, oversight and

	governance signal for review.”	audit trail.”
Why fairness monitoring remains	Consumer-harm prevention, model-risk governance, reputational risk, proxy-discrimination detection, and readiness for state or other legal regimes.	Required by high-risk AI governance logic and fundamental-rights risk management.

Suggested Rule-Pack Implementation

The key implementation change is to split “legal findings” from “fairness monitoring”. The tool can still compute the same metrics in both jurisdictions, but the US mode should not use those metrics as a legal status trigger unless they are connected to prohibited-basis treatment, proxy intent, override patterns or adverse-action defects.

```
JURISDICTION_RULES = {
    "US": {
        "name": "United States",
        "regulatory_focus": (
            "ECOA / Regulation B intentional discrimination, "
            "disparate-treatment review, adverse-action explainability, "
            "and proxy-risk governance"
        ),
        "legal_theory": "intent_based_disparate_treatment",
        "uses_disparate_impact_as_legal_trigger": False,
        "requires_adverse_action_reasons": True,
        "requires_specific_reason_codes": True,
        "requires_prohibited_basis_feature_check": True,
        "requires_similarly_situated_comparison": True,
        "requires_override_pattern_review": True,
        "requires_proxy_risk_review": True,
        "outcome_metrics_role": "monitoring_only_not_legal_violation",
        "monitoring_metrics": [
            "approval_rate_by_group",
            "demographic_parity_difference",
            "false_negative_rate_by_group",
            "calibration_by_group",
            "proxy_correlation_with_protected_attributes",
            "model_drift_by_group"
        ],
    },
    "EU": {
        "name": "European Union",
        "regulatory_focus": (
            "EU AI Act high-risk AI governance for creditworthiness "
            "and credit-score systems"
        ),
        "legal_theory": "high_risk_ai_outcome_and_governance_controls",
        "uses_disparate_impact_as_legal_trigger": True,
        "requires_bias_testing": True,
        "requires_technical_documentation": True,
        "requires_human_oversight_record": True,
        "requires_audit_log": True,
        "requires_lifecycle_monitoring": True,
        "outcome_metrics_role": "high_risk_ai_governance_evidence",
    },
}
```

Revised Status Logic

```
def build_us_report(scored_df, model, rules):
    legal_findings = []
    monitoring_findings = []

    # Legal checks: these can trigger US legal review.
    legal_findings += check_direct_use_of_protected_attributes(model)
    legal_findings += check_similarly_situated_differential_treatment(scored_df)
    legal_findings += check_human_override_patterns(scored_df)
    legal_findings += check_adverse_action_reason_specificity(scored_df, model)
    legal_findings += check_proxy_risk_with_intent_or_pretext_indicators(scored_df)

    # Monitoring-only checks: these do not establish a US legal violation by themselves.
    monitoring_findings += compute_group_outcome_metrics(scored_df)
    monitoring_findings += compute_false_negative_rate_gaps(scored_df)
    monitoring_findings += compute_proxy_correlation_metrics(scored_df)

    if legal_findings:
        status = "US_LEGAL_REVIEW_REQUIRED"
    elif monitoring_findings:
        status = "US_MONITORING_REVIEW_RECOMMENDED"
    else:
        status = "NO_US_LEGAL_FINDING"

    return {
        "status": status,
        "legal_findings": legal_findings,
        "monitoring_only_fairness_signals": monitoring_findings,
        "disclaimer": (
            "Outcome disparity alone is not treated as a US ECOA violation. "
            "It is retained as an internal governance and proxy-risk signal."
        )
    }
```

How the Answer Changes the Interpretation of the Prototype Result

In the existing run, the ethnicity disparate-impact ratio was 0.6575, so both US and EU reports returned REVIEW_REQUIRED. After the redesign, the same metric would still be calculated in both jurisdictions, but the labels would diverge more clearly:

Same metric	US redesigned interpretation	EU interpretation
Ethnicity disparate-impact ratio = 0.6575	Monitoring review recommended. The report would say this is not, by itself, an ECOA legal violation. It may indicate proxy risk, model-risk concern, reputational risk or the need to inspect similarly situated applicants and override records.	High-risk governance review required. The disparity remains part of the AI Act high-risk system evidence package and should trigger bias mitigation, technical documentation, human oversight and audit logging.

Why I Would Not Turn Off US Fairness Monitoring

First, a RegTech tool should support risk governance, not merely reproduce the minimum legal liability test. Even if disparate impact is not used as the US legal standard, large unexplained group disparities can still reveal model-risk problems, weak data quality, reputational exposure or consumer-harm risk.

Second, outcome metrics help detect proxy risk. A model may not directly use gender or ethnicity, but variables such as income, geography, employment type or credit history can still produce suspicious group patterns. In US mode, the tool should use those patterns to ask whether there is evidence of intentional proxy use, pretext, override inconsistency or similarly situated differential treatment.

Third, the same bank may face state, local, investor, prudential, reputational or internal model-risk expectations that are broader than the narrow ECOA liability trigger. Keeping monitoring metrics allows the tool to remain useful to the CCO without overstating the law.

Fourth, this design is more honest. It avoids saying “no fairness issue” simply because US disparate-impact liability is removed. Instead, it says: “no US legal finding from the metric alone, but governance review may still be prudent.”

Final Response

My revised answer would therefore be: I would not turn off fairness monitoring in the US pack. I would decouple fairness metrics from legal liability. In US mode, disparate-impact metrics would be retained as monitoring-only governance signals that may trigger proxy-risk, override-pattern or adverse-action-quality review, but they would not automatically create a US ECOA violation label. The US legal status would be driven by direct prohibited-basis use, disparate treatment, pretext, discouragement, inaccurate adverse-action reasons and suspicious proxy or override evidence. In EU mode, by contrast, the same outcome metrics would remain compliance-relevant because creditworthiness and credit-score AI systems are high-risk AI systems under the AI Act. This redesign makes the tool more genuinely jurisdiction-aware: the same metric can be calculated in both jurisdictions, but the legal meaning, escalation path and evidence package are different.

References

[1] Consumer Financial Protection Bureau, Equal Credit Opportunity Act (Regulation B), Final Rule, 2026. The final rule states that ECOA does not authorize disparate-impact liability and provides the effective-date framework. https://public-inspection.federalregister.gov/2026-07804.pdf

[2] Consumer Financial Protection Bureau, "Providing equal credit opportunities (ECOA)," updated 2026, noting the April 22, 2026 final rule amending Regulation B provisions related to disparate impact, discouragement and SPCPs. https://www.consumerfinance.gov/compliance/compliance-resources/other-applicable-requirements/equal-credit-opportunity-act/

[3] Consumer Financial Protection Bureau, 12 CFR § 1002.4, Regulation B General Rules, current regulation and official interpretation on prohibited-basis discrimination and disparate treatment. https://www.consumerfinance.gov/rules-policy/regulations/1002/4/

[4] Consumer Financial Protection Bureau, Circular 2022-03, adverse-action notification requirements for credit decisions based on complex algorithms. https://www.consumerfinance.gov/compliance/circulars/circular-2022-03-adverse-action-notification-requirements-in-connection-with-credit-decisions-based-on-complex-algorithms/

[5] European Commission AI Act Service Desk, Annex III, high-risk AI systems, including AI systems used to evaluate creditworthiness or establish credit scores. https://ai-act-service-desk.ec.europa.eu/en/ai-act/annex-3

[6] European Commission, "AI Act | Shaping Europe's digital future," describing the AI Act as a risk-based framework for trustworthy AI. https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai